THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	Common prime factors give HCF $= 6$
2	**(d)** 6	Substitute $x = 4$: $p(4) = 6$
3	**(c)** infinitely many solutions (coincident lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$
4	**(c)** $x^2 + 3x + 2 = 0$	Degree must be 2
5	**(b)** 4	$a_2 - a_1 = 4$
6	**(c)** 5	$\sqrt{3^2 + 4^2} = 5$
7	**(c)** 5	$\sqrt{x^2 + y^2}$
8	**(d)** $\frac{4}{5}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
9	**(a)** 7 m	$h = 7\tan 45^\circ = 7 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 7$ m
10	**(a)** 4 cm	$PT^2 = OP^2 - OT^2 = 5^2 - 3^2 = 16$; $PT = 4$ cm
11	**(d)** 154 cm²	$\frac{22}{7} \times 7 \times 7$; Area $= 154$ cm^2
12	**(c)** 220 cm²	$2 \times \frac{22}{7} \times 7 \times 5$; CSA $= 220$ cm^2
13	**(a)** 15	$\frac{10+20}{2}$; Class mark $= 15$
14	**(d)** $\frac{1}{2}$	Favourable outcomes: 2, 4, 6; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
15	**(c)** 12	LCM $= \dfrac{4 \times 6}{2} = 12$
16	**(c)** 4	$2x - 8 = 0 \Rightarrow x = 4$
17	**(d)** $(3, 2)$	$2(3) + 3(2) = 12 = 12$
18	**(b)** 2	At $x = 2$ the left side equals 0, so 2 is a root
19	**(b)** 39	$a + (n-1)d = 39$
20	**(a)** 3	Perimeter ratio = side ratio
21	**(c)** $(4,\ 6)$	Average the coordinates
22	**(a)** $\frac{4}{3}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
23	**(c)** $5\sqrt{3}$ m	$h = 5\tan 60^\circ = 5 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 5\sqrt{3}$ m
24	**(d)** 6 cm	$r^2 = OP^2 - PT^2 = 10^2 - 8^2 = 36$; $r = 6$ cm
25	**(a)** 44 cm	$2 \times \frac{22}{7} \times 7$; Circumference $= 44$ cm
26	**(a)** 1540 cm³	$\frac{22}{7} \times 7^2 \times 10$; Volume $= 1540$ cm^3
27	**(c)** 30–40	It belongs to the class 30–40; Modal class $= 30$–40
28	**(d)** $\frac{1}{3}$	Favourable outcomes: 3, 6; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
29	**(c)** 3	$12 = 2^2 \times 3$
30	**(b)** 5	Sum of zeroes $= -b/a$
31	**(a)** 7	$k = 2(2) + 3(1)$
32	**(d)** -3	Compare with $ax^2 + bx + c = 0$
33	**(a)** 55	$S_n = \frac{n}{2}[2a + (n-1)d]$
34	**(d)** 6	$\frac{AD}{DB} = \frac{AE}{EC}$
35	**(a)** first quadrant	Signs of coordinates
36	**(a)** $\frac{1}{2}$	Standard value
37	**(b)** $3\sqrt{3}$ m	$h = 9\tan 30^\circ = 9 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 3\sqrt{3}$ m
38	**(b)** 25 cm	$OP^2 = r^2 + PT^2 = 7^2 + 24^2 = 625$; $OP = 25$ cm
39	**(c)** 7 cm	$r = \frac{44 \times 7}{2 \times 22}$; $r = 7$ cm
40	**(b)** 25 cm	$l^2 = 7^2 + 24^2 = 625$; $l = 25$ cm

41	**(c)** 40–50	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class $= 40$–50
42	**(a)** $\frac{1}{4}$	Favourable outcomes: HH; number $= 1$; $P = \frac{1}{4}$
43	**(a)** $\sqrt{7}$	7 is not a perfect square, so $\sqrt{7}$ is irrational
44	**(b)** 6	Product of zeroes $= c/a$
45	**(b)** 7	$x + y = 10$, $x - y = 4$ gives $2x = 14$
46	**(b)** 2 and 3	Put each factor equal to zero
47	**(d)** $4, 7, 10, 13, \ldots$	Only $4, 7, 10, 13, \ldots$ has the same difference 3 between consecutive terms
48	**(d)** 3	Side ratio is the square root of the area ratio
49	**(b)** $(3,\ 0)$	$y = 0$ on the x-axis
50	**(c)** 1	Identity

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** 9 m	$x = 9\cot 45^\circ$; Shadow $= 9$ m
2	**(d)** 14 cm	Perimeter $= PA + AO + OB + BP = 2(4) + 2(3)$; Perimeter $= 14$ cm
3	**(a)** 7 cm	$r^2 = \frac{154\times 7}{22} = 49$; $r = 7$ cm
4	**(c)** 550 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm^2
5	**(b)** 50	$3 + 9 + 4 + 18 + 16$; $\sum f_i = 50$
6	**(a)** $\frac{3}{8}$	Favourable balls $= 3$; $P = \frac{3}{8}$
7	**(c)** $2^2 \times 3 \times 5$	Divide repeatedly by primes: $60 = 2^2 \times 3 \times 5$
8	**(c)** 2	A polynomial of degree n has at most n zeroes
9	**(b)** y-axis	All points have the same x-coordinate
10	**(c)** 1	$D = b^2 - 4ac$
11	**(a)** 8	$a_2 = a + d$
12	**(b)** 5	Pythagoras
13	**(d)** 10	Distance formula
14	**(b)** 65°	$\sin\theta = \cos(90^\circ - \theta)$
15	**(b)** 12 m	$\tan 45^\circ = \frac{12}{x}$ so $x = 12\cot 45^\circ$; Distance $= 12$ m
16	**(c)** 70°	In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 110^\circ$; $\angle APB = 70^\circ$
17	**(d)** 231 cm²	$\frac{60}{360} \times \frac{22}{7} \times 21^2$; Area $= 231$ cm^2
18	**(b)** 616 cm²	$4 \times \frac{22}{7} \times 7^2$; Surface area $= 616$ cm^2
19	**(c)** 12	$3 + 9$; $cf = 12$
20	**(d)** $\frac{3}{10}$	Favourable numbers: 3, 6, 9; number $= 3$; $P = \frac{3}{10}$
21	**(b)** terminating	Denominator 8: only 2 and 5 as prime factors
22	**(b)** 2	$p(2) = 0 = 0$, while the other values do not make $p(x)$ zero
23	**(a)** coincident	One equation is a multiple of the other
24	**(a)** two distinct real roots	$D = b^2 - 4ac = 1$
25	**(b)** 3	$a_3 = a + 2d$
26	**(b)** equal	AAA similarity
27	**(d)** 5	Abscissa is the x-coordinate
28	**(d)** $\cos\theta$	$\sec\theta = 1/\cos\theta$
29	**(d)** $4\sqrt{3}$ m	Height $= 8\sin 60^\circ = 8 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 4\sqrt{3}$ m
30	**(d)** 60°	$\angle AOP = 180^\circ - 90^\circ - 30^\circ$; $\angle AOP = 60^\circ$
31	**(d)** 22 cm	$\frac{60}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 22$ cm
32	**(c)** 38808 cm³	$\frac{4}{3} \times \frac{22}{7} \times 21^3$; Volume $= 38808$ cm^3
33	**(a)** 34.8	$\sum f_i = 50$ and $\sum f_i x_i = 1740$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$
34	**(d)** $\frac{3}{4}$	$P(\text{not } E) = 1 - \frac{1}{4}$; $P(\text{not } E) = \frac{3}{4}$
35	**(d)** 10	Common prime factors give HCF $= 10$
36	**(a)** 16	Substitute $x = 3$: $p(3) = 16$
37	**(a)** no solution (parallel lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{5}{7}$
38	**(c)** $x^2 - 5x + 6 = 0$	Degree must be 2
39	**(b)** -3	$a_2 - a_1 = -3$
40	**(d)** 13	$\sqrt{5^2 + 12^2} = 13$
41	**(b)** 13	$\sqrt{x^2 + y^2}$

42	**(a)** $\frac{12}{13}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
43	**(c)** line of sight	Definition
44	**(a)** one point	Definition of a tangent
45	**(a)** 36 cm	Arc $= \pi r = 22$ cm; diameter $= 14$ cm; Perimeter $= 36$ cm
46	**(c)** 462 cm²	$3 \times \frac{22}{7} \times 7^2$; TSA $= 462$ cm^2
47	**(b)** 58.75	$l = 50$, $f_1 = 18$, $f_0 = 4$, $f_2 = 16$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{14}{16} \times 10 = 58.75$
48	**(d)** 1	A sure event always happens
49	**(d)** 18	LCM $= \dfrac{6 \times 9}{3} = 18$
50	**(c)** -3	$3x + 9 = 0 \Rightarrow x = -3$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** $(2, 5)$	$3(2) + 1(5) = 11 = 11$
2	**(a)** 3	At $x = 3$ the left side equals 0, so 3 is a root
3	**(c)** 26	$a + (n-1)d = 26$
4	**(b)** 5	Perimeter ratio = side ratio
5	**(a)** $(4,\ 3)$	Average the coordinates
6	**(b)** $\frac{12}{5}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
7	**(a)** 12 m	$h = 12\tan 45^\circ = 12 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = 12 m
8	**(c)** 8 cm	$PT^2 = OP^2 - OT^2 = 10^2 - 6^2 = 64$; $PT = 8$ cm
9	**(c)** 25 cm	Quarter arc $= \frac{\pi r}{2} = 11$ cm; $2r = 14$ cm; Perimeter $= 25$ cm
10	**(b)** 880 cm²	$2 \times \frac{22}{7} \times 14 \times 10$; CSA $= 880$ cm^2
11	**(d)** 21	Median class 20–30: $l = 20$, $cf = 12$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{12.5 - 12}{5} \times 10 = 21$
12	**(c)** $\frac{1}{2}$	Favourable outcomes: 1, 3, 5; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
13	**(b)** 3	$30 = 2 \times 3 \times 5$
14	**(a)** 4	Sum of zeroes $= -b/a$
15	**(b)** 14	$k = 3(4) + 1(2)$
16	**(d)** 7	Compare with $ax^2 + bx + c = 0$
17	**(d)** 72	$S_n = \frac{n}{2}[2a + (n-1)d]$
18	**(c)** 8	$\frac{AD}{DB} = \frac{AE}{EC}$
19	**(d)** second quadrant	Signs of coordinates
20	**(d)** $\frac{1}{2}$	Standard value
21	**(a)** $8\sqrt{3}$ m	$h = 8\tan 60^\circ = 8 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 8\sqrt{3}$ m
22	**(c)** 5 cm	$r^2 = OP^2 - PT^2 = 13^2 - 12^2 = 25$; $r = 5$ cm
23	**(b)** 616 cm²	$\frac{22}{7} \times 14 \times 14$; Area $= 616$ cm^2
24	**(b)** 3080 cm³	$\frac{22}{7} \times 14^2 \times 5$; Volume $= 3080$ cm^3
25	**(a)** 30	Class limits
26	**(a)** $\frac{1}{3}$	Favourable outcomes: 1, 4; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
27	**(c)** $\sqrt{11}$	11 is not a perfect square, so $\sqrt{11}$ is irrational
28	**(d)** -5	Product of zeroes $= c/a$
29	**(b)** 10	$x + y = 14$, $x - y = 6$ gives $2x = 20$
30	**(c)** -1 and 4	Put each factor equal to zero
31	**(b)** $9, 7, 5, 3, \ldots$	Only $9, 7, 5, 3, \ldots$ has the same difference -2 between consecutive terms
32	**(d)** 5	Side ratio is the square root of the area ratio
33	**(d)** $(-4,\ 0)$	$y = 0$ on the x-axis
34	**(d)** 1	Identity
35	**(d)** $4\sqrt{3}$ m	$h = 12\tan 30^\circ = 12 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 4\sqrt{3}$ m
36	**(b)** 25 cm	$OP^2 = r^2 + PT^2 = 15^2 + 20^2 = 625$; $OP = 25$ cm
37	**(b)** 88 cm	$2 \times \frac{22}{7} \times 14$; Circumference $= 88$ cm
38	**(a)** 50 cm	$l^2 = 14^2 + 48^2 = 2500$; $l = 50$ cm
39	**(b)** 25	$\frac{20+30}{2}$; Class mark $= 25$
40	**(b)** $\frac{3}{4}$	Favourable outcomes: HH, HT, TH; number $= 3$; $P = \frac{3}{4}$

41	**(c)** $2^2 \times 3 \times 7$	Divide repeatedly by primes: $84 = 2^2 \times 3 \times 7$
42	**(c)** 3	A polynomial of degree n has at most n zeroes
43	**(b)** y-axis	All points have the same x-coordinate
44	**(b)** 1	$D = b^2 - 4ac$
45	**(c)** 6	$a_2 = a + d$
46	**(b)** 13	Pythagoras
47	**(a)** 13	Distance formula
48	**(d)** 50°	$\sin\theta = \cos(90^\circ - \theta)$
49	**(d)** 15 m	$x = 15\cot 45^\circ$; Shadow $= 15$ m
50	**(a)** 28 cm	Perimeter $= PA + AO + OB + BP = 2(8) + 2(6)$; Perimeter $= 28$ cm

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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